

DERMAI: CLINICAL DERMATOLOGY ACQUISITION THROUGH QUALITY-DRIVEN IMAGE COLLECTION FOR AI CLASSIFICATION IN MOBILE

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ABSTRACT

AI-based dermatology adoption remains limited by biased datasets, variable image quality, and limited validation. We introduce DermAI, a lightweight, smartphone-based application that enables real-time capture, annotation, and classification of skin lesions during routine consultations. Unlike prior dermoscopy-focused tools, DermAI performs on-device quality checks, and local model adaptation. The DermAI clinical dataset, encompasses a wide range of skin tones, ethnicity and source devices. In preliminary experiments, models trained on public datasets failed to generalize to our samples, while fine-tuning with local data improved performance. These results highlight the importance of standardized, diverse data collection aligned with healthcare needs and oriented to machine learning development. The dataset will be made available for research purposes upon request to the corresponding authors.

Index Terms— Dermatology application, Skin lesion, Clinical dataset

1. INTRODUCTION

The development of AI in healthcare has rapidly increased, particularly in medical image analysis [1]. In dermatology, such models can support clinicians by providing preliminary lesion assessments [2], [3]. However, despite the availability of several public datasets [4], [5], [6], many suffer from limited diversity, lack of acquisition standards, and imaging artifacts, which reduce their clinical applicability [7]. Moreover, while these datasets consider clinical challenges, they rarely address the practical constraints required for machine learning pipelines. Consequently, the models trained often fail to generalize to real-world conditions, especially in public healthcare systems serving diverse populations.

To address these limitations, we introduce DermAI, designed for standardized dermatological image acquisition and oriented towards AI training in clinical environments. The platform enforces consistent capture and performs real-time

quality checks. Tailored to dermatologists and general practitioners, it facilitates early triage and risk assessment while generating clinically meaningful training data. Using the platform, we constructed a new clinical dataset collected during routine consultations at a Brazilian public university hospital, encompassing a wide range of skin tones, ethnicities, devices, and environmental conditions. We benchmarked convolutional neural networks on this dataset and compared their performance to models trained on public datasets. Models trained on public data showed limited generalization under real clinical variability. This result emphasizes the necessity of context-specific samples and demonstrates DermAI as a promising practical tool.

2. AVAILABLE DATASETS

Several image-based dermatology datasets have supported the development of learning-based skin lesion classification; Typically categorized into clinical [4], [5] and dermoscopic [6]. However, they present critical limitations [1], [4], including restricted access, unclear acquisition protocols, and underrepresentation of diverse skin tones and ethnic groups, hindering fairness and model generalization [8]. The ISIC Archive¹ remains the largest dermoscopic repository with histopathologically confirmed samples, yet is predominantly composed of lighter skin types [7]. Clinical datasets such as DDI [4] and PAD-UFES-20 [5] improve diversity but still contain acquisition artifacts, limited variability across environments, and provide relatively few samples.

3. DERMAI

DermAI is a scalable solution designed for dermatological workflows. It standardizes image acquisition, particularly in low-resource settings where non-specialist users rely on standard mobile devices. Thus, DermAI enables reproducible and controlled image capture suitable for machine learning training under diverse clinical conditions.

¹<https://www.isic-archive.com/>

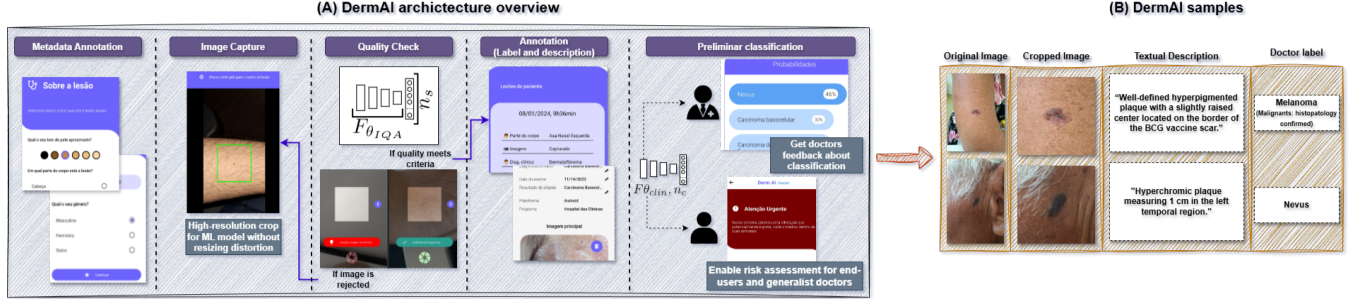


Fig. 1. An overview of the APP structure. (A) Shows the main steps: image capture, quality validation, annotation, label review, and classification. (B) Illustrate the resulting samples: acquired images, textual description, and initial label.

3.1. Acquisition, quality validation and label review

We collaborated with board-certified dermatologists, residents, and medical students to collect clinical data during consultations at the *Hospital das Clínicas, Universidade Federal de Pernambuco*. Each patient received a unique identifier to link images and metadata. A standardized form was completed for every case, including patient age, Fitzpatrick skin phototype, gender, and lesion location. Lesion images were captured directly using smartphone cameras and paired with short clinical descriptions, producing image-text annotations (Figure 1-b). Device attributes were also recorded (model, OS, camera specifications). Medical students accompanied clinicians during consultations but were responsible only for image acquisition. All samples were reviewed and validated by the attending dermatologist, and the preliminary diagnosis reflects their clinical judgment. To address common issues in clinical dermatology imaging (e.g., ink markings, gel artifacts, or glare), the application provides specific guidelines; Clinicians should avoid pre-marking, center the lesion using on-screen guides, and capture images at a standardized distance of approximately 5 cm. Environmental conditions remain unconstrained, reflecting clinical variability. To ensure patient privacy, images are acquired without faces or any identifiable personal characteristics and are cropped to focus exclusively on the lesion. No personal or identifying information is recorded. All images are securely stored on internal servers and are accessible only to authorized administrators.

Clinical image acquisition is naturally affected by uncontrolled factors, such as ambient lighting, camera jitter, inconsistent distances, angles, or focus. To mitigate these effects, we preprocess images by cropping a central region and enforcing a square aspect ratio, which reduces background content, emphasizes the lesion, and minimizes distortion. During acquisition, a preview of the cropped region helps to ensure framing consistency. This design aligns with common machine learning pipelines that require square inputs, limiting geometric distortion from the point of capture. Both the original and cropped images are stored. We also developed a learning-based image quality assessment (IQA) model

based on a MobileNetV3 backbone ($F_{\theta_{IQA}}$), followed by a linear layer with SiLU activation, dropout (0.2), and a sigmoid output. The model predicts $n_s = 4$ binary quality indicators corresponding to sharpness, blur, exposure (over/under), and compression artifacts. It was pretrained on natural images and subsequently fine-tuned on domain-specific dermatological images, with severe distortions simulated through data augmentation. The applied augmentations include Gaussian blur, motion blur, focal blur, brightness variations, and noise, reflecting artifacts commonly observed in smartphone-acquired images. Training was performed using the Adam optimizer with a learning rate of 10^{-3} , minimizing a multi-label binary cross-entropy loss (Equation 1).

$$\mathcal{L}_{IQA} = - \sum_{i=1}^4 [y_i \log(\hat{y}_i) + (1 - y_i) \log(1 - \hat{y}_i)] \quad (1)$$

Rather than correcting the images, it quantifies image quality and prompts the user to recapture in real time, preventing severely degraded samples from entering the dataset. After acquisition, samples that meet quality criteria are reviewed by dermatologists, who mark the lesion center and define a circular region of interest. Each image contains a single lesion; multiple lesions require separate captures to ensure consistent annotation. Clinicians may flag lesions suspected of malignancy, prompting biopsy follow-up and later inclusion of histopathological results. This workflow ensures that malignant-suspect cases undergo systematic verification, improving the reliability of diagnostic labels in the dataset.

3.2. Dataset

At this time, DermAI data collection had been ongoing for two years and comprised 3,401 images (2,273 benign, 608 malignant, and 520 pre-malignant). It includes common lesion types, such as melanoma (30), basal cell carcinoma (477), squamous cell carcinoma (142), nevus (568), actinic keratosis (520), benign keratosis (706), and solar lentigo (285), among others. Figure 2 presents representative samples and comparisons with existing datasets, highlighting re-

duced imaging artifacts due to standardized acquisition. Data collection continues, and the dataset is expected to grow.



Fig. 2. Samples collected with DermAI (top row), illustrating artifact-controlled acquisition. For comparison, the bottom row shows samples from public sources, which may include distractors, blur, or overexposure.

3.3. Preliminary classification and medical feedback

Finally, a classification model provides a preliminary suggestion. We use a CNN-based model, following prior work [4], [5]. Several backbones were evaluated, and the best-performing ones were converted for local on-device inference. The model classifies n_c lesion categories to support clinical decision-making and gather feedback from dermatologists. For general practitioners or non-specialist users, the binary risk assessment (benign vs. potentially malignant) is provided. The CNN architecture is followed by an average pooling and a fully connected layer trained with categorical cross-entropy. The clinicians review the predictions, confirming accordance or indicating uncertainty and providing their own diagnostic hypothesis. This feedback enables performance monitoring under real clinical use. All lesion annotations and metadata are retained; lesions flagged as malignant-suspect are prioritized for biopsy follow-up. A supervisory interface enables board-certified dermatologists review annotation consistency.

Additionally, we evaluate an ensemble strategy. Let $\{f_1, f_2, \dots, f_K\}$ denote K trained classifiers, each producing a probability vector $p_k \in \mathbf{R}^{n_c}$. We consider two schemes: (i) **Majority Vote** (Eq. 2), which emphasizes consensus among models and reduces variance due to classifier-specific biases; (ii) **Learned Fusion** a MLP-based ($g_\phi(\cdot)$) that receives the concatenated model outputs (Eq. 3) and learns the correlations between classifier confidence patterns. The MLP consists of two fully connected layers with ReLU activation and dropout, trained using categorical cross-entropy. These ensemble strategies improve predictive stability and sensitivity, particularly for malignant-suspect cases, by leveraging complementary decision patterns among architectures.

$$\hat{y} = \arg \max_c \sum_{k=1}^K \mathbf{1}(\arg \max(p_k) = c) \quad (2)$$

$$\hat{y} = \arg \max g_\phi([p_1 \| p_2 \| \dots \| p_K]) \quad (3)$$

4. EXPERIMENTS AND DISCUSSIONS

We trained multiple CNN architectures (Table 1) on the PAD-UFES-20 (PAD) dataset [5], replicating the original setup while adding recent backbones. Since PAD does not define splits, we established fixed training and test sets (80/10/10). We then evaluated the resulting models on DDI [4] and DermAI to assess out-of-distribution generalization. We focused on lightweight models suitable for on-device execution. The DDI, due to its small size, was used mainly for testing. Evaluation was restricted to the six classes in PAD (melanoma, nevus, seborrheic keratosis, actinic keratosis, basal cell carcinoma, and squamous cell carcinoma), allowing direct analysis between the three datasets. Performance was measured using accuracy (ACC), recall (R), precision (P), and F1-score (F1) [5]. As shown in Table 1, PAD-trained models align with previous results but show limited generalization when tested on DDI and DermAI, particularly in recall, which is critical for minimizing missed malignant cases. Although PAD is visually diverse, it contains acquisition artifacts (e.g., ink markings, blur) and a limited skin tones variety, which constrain robust feature learning. Retraining the same architectures using DermAI improved cross-dataset performance, reflecting the benefits of controlled and diverse acquisition. While it does not achieve the highest scores in every scenario, the model maintains stable scores on PAD and marginally improves on DDI, the most challenging dataset due to its high noise and variability. We further applied DermAI-based quality filtering to PAD (PAD_filter) and retrained the models. Despite using fewer samples, the PAD_filter improved performance, highlighting the importance of curated image features. Combining PAD_filter with DermAI yielded the best overall results. Adding DDI samples to training, in addition to improving scores on the same, did not improve overall performance, likely due to their scarcity and noise, reinforcing that increasing data quantity alone is insufficient.

5. CONCLUSION

We demonstrate that clinical algorithms often face significant limitations due to domain variation, exposing weaknesses across heterogeneous data sources. Consequently, such models cannot be fully trusted in practical clinical use. Thus, we introduce DermAI, an acquisition framework that ensures high-quality image capture while remaining accessible for diverse acquisition scenarios. DermAI addresses limited diversity, inconsistent image quality, and standard acquisition structured for machine learning. It enables datasets to grow continuously, supporting model refinement by incorporating other sources. Being model-agnostic, DermAI fosters reproducible research and adaptation, laying the groundwork for AI-assisted dermatological care.

Table 1. Cross-dataset classification performance across architectures evaluated using accuracy (ACC), recall (R), precision (P), and F1-score (F1). The first eight models are trained on PAD-UFES-20 (PAD), and the subsequent eight on DermAI. All models are evaluated on three target datasets. The final five rows report the mean performance across architectures trained on: (i) PAD, (ii) PAD (filtered), (iii) DermAI, (iv) combining PAD (filtered) + DermAI, and (v) PAD + DermAI + DDI.

	PAD-UFES-20 [5]				DDI [4]				DermAI			
	ACC	R	P	F1	ACC	R	P	F1	ACC	R	P	F1
DenseNet (PAD)	0.7533	0.6198	0.6343	0.6280	0.1987	0.1949	0.1270	0.1537	0.4648	0.5583	0.5745	0.5531
ConvNeXt (PAD)	0.7503	0.6523	0.7144	0.6718	0.1862	0.1768	0.1518	0.1180	0.4794	0.5719	0.5906	0.5756
MobileNet v3 (PAD)	0.7050	0.5383	0.6824	0.6018	0.1829	0.1408	0.0568	0.0644	0.4361	0.4366	0.4436	0.4300
EfficientNet v2 (PAD)	0.7554	0.6523	0.7144	0.6718	0.1768	0.1518	0.1060	0.1180	0.4897	0.4824	0.4821	0.4968
Resnet (PAD)	0.7073	0.7072	0.6809	0.6883	0.1280	0.1156	0.0438	0.0590	0.5568	0.4824	0.4968	0.4821
Ensemble-major (PAD)	0.7692	0.7400	0.7925	0.7653	0.1737	0.1105	0.0383	0.0517	0.5274	0.4840	0.5511	0.5153
Ensemble-fusion (PAD)	0.7795	0.7800	0.8000	0.7898	0.1253	0.1167	0.0842	0.0978	0.5283	0.5466	0.5161	0.5309
DenseNet (DermAI)	0.7317	0.7386	0.7240	0.7224	0.3467	0.3295	0.2933	0.3103	0.8209	0.8176	0.8209	0.8192
ConvNeXt (DermAI)	0.7526	0.7317	0.7201	0.7205	0.3953	0.3949	0.3987	0.3867	0.8102	0.8092	0.8102	0.8096
MobileNet v3 (DermAI)	0.6317	0.6317	0.6201	0.6205	0.2927	0.1899	0.2077	0.1984	0.7834	0.7788	0.7834	0.7765
EfficientNet v2 (DermAI)	0.6247	0.6247	0.6233	0.6206	0.3943	0.3383	0.3832	0.3593	0.7567	0.7497	0.7567	0.7531
Resnet (DermAI)	0.6993	0.6933	0.6835	0.6687	0.5050	0.3824	0.4383	0.4008	0.7834	0.7805	0.7834	0.7819
Ensemble-major (DermAI)	0.7902	0.7939	0.7902	0.7910	0.4807	0.3156	0.2949	0.3048	0.8234	0.8234	0.8338	0.8232
Ensemble-fusion (DermAI)	0.8035	0.8035	0.8033	0.8033	0.4903	0.4880	0.4999	0.4938	0.8467	0.8583	0.8560	0.8571
AVG-all models (PAD)	0.7457	0.6700	0.7170	0.6881	0.1673	0.1439	0.0873	0.0941	0.4975	0.5088	0.5221	0.5119
AVG-all models (PAD.filter)	0.7817	0.7812	0.7817	0.7850	0.1583	0.0925	0.0225	0.03601	0.5196	0.5291	0.5278	0.5217
AVG-all models (DermAI)	0.7184	0.7168	0.7092	0.7067	0.4150	0.3483	0.3594	0.3505	0.8035	0.8025	0.8063	0.8088
AVG-all models (PAD.filter + DermAI)	0.8505	0.8462	0.8803	0.8607	0.5698	0.4693	0.5793	0.5185	0.8511	0.8420	0.8754	0.8583
AVG-all models (PAD + DermAI + DDI)	0.7971	0.7971	0.8158	0.7961	0.7778	0.7263	0.7311	0.7286	0.7956	0.8089	0.7956	0.7952

6. COMPLIANCE WITH ETHICAL STANDARDS

The study involving human participants was reviewed and approved by the Research Ethics Committee of the Federal University of Pernambuco (CEP-UFPE), which operates within the national CEP/CONEP system via the Plataforma Brasil (CAAE ID 75991523.3.0000.8807). The research was conducted in accordance with the ethical principles embodied in the Declaration of Helsinki and consistent with the Brazilian National Health Council's Resolution 466/12 (and all applicable local statutory requirements). Written informed consent to participate in this study was obtained from all participants.

7. ACKNOWLEDGMENTS

This work was partially supported by INES.IA (National Institute of Science and Technology for Software Engineering Based on and for Artificial Intelligence) www.ines.org.br, CNPq grant 408817/2024-0. The project was supported by the Ministry of Science, Technology, and Innovation of Brazil, with resources from Law No. 8,248, dated October 23, 1991, under the scope of the PPI-SOFTEX, coordinated by Softex and published under RESIDÊNCIA EM TIC 63 – ROBÓTICA E IA – FASE II, DOU 23076.043130/2025-27.

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